

Curriculum Vitae - Li Hongmin

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Primary Affiliation: Sato Laboratory for Biomedical Data Science, Institute of Science Tokyo (Laboratory)

Guest Affiliation: Frith Lab, The University of Tokyo (Laboratory)

Research Interests

- AI-Automated Scientific Workflows
 - AI for Science and Scientific Discovery
 - AI Research and Evaluation
 - Bioinformatics and Biological Sequence Analysis
 - Biomolecular Sequence Design and Optimization
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Education

- **2019.04 —2022.03 | Ph.D. in Computer Science (Information Systems Engineering)** —University of Tsukuba
 - **2017.04 —2019.03 | Master's in Computer Science (Information Systems Engineering)** —University of Tsukuba
 - **2011.09 —2015.07 | Bachelor's in Electronic Information Engineering** —Ningxia University
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Academic & industry appointments

- **2026.4 —Present | Researcher** —Sato Laboratory for Biomedical Data Science, Institute of Science Tokyo; Laboratory
 - **2026.4 —Present | Guest Researcher** —Frith Lab, The University of Tokyo; Laboratory
 - **2026.6 —Present | Founder** —AlphaScience Lab (Research community); Website
 - **2023.5 —2026.3 | Postdoctoral Researcher** —The University of Tokyo
 - **2022.10 —2023.5 | Machine Learning Engineer** —HAOMO.AI —Autonomous-driving project · 蓝色空间领航者 (A super high salary project)
 - **2022.4 —2022.10 | Postdoctoral Researcher** —The University of Tokyo; Details
 - **2021.7 —2021.9 | Machine Learning Engineer (Internship)** —HAOMO.AI
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Research funding & computing support

- **2026.06 | Google Cloud TPU Builders Award** —Cloud computing support · USD 7,500 in Google Cloud computing credits
 - **2026.04 | Grant-in-Aid for Early-Career Scientists (KAKENHI)** —Development of an Input Data Differentiable Integrated Framework (ID3) for Biomolecular Sequence Design · ¥4,550,000; Project info
 - **2025.10 | Google research support** —Biological sequence optimization · USD 30,000
 - **2024.4 | Grant-in-Aid for Early-Career Scientists (KAKENHI)** —Development of a Large-Scale Language Model Integrating RNA Sequences and Text · ¥4,420,000; Project info
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Publications

Books

- **2026.08** | Hongmin Li. “*Academic Collapse Does Not Happen All at Once: A Conditional Theory of Epistemic Cascades, Expertise Cliffs, and Institutional Recoupling*” . Book · Independently published · 310 pp. · ISBN 979-8192323182. Amazon

Preprints

- **2026.09** | Hongmin Li, Wanli Zhao. “*Where Scientific Search Agents Fail: Decision-Checkpoint Auditing of Exposure and Inspection Attempts*” . arXiv preprint · 2609.38670. arXiv
- **2026.09** | Hongmin Li. “*The Hodge Conjecture for Odd-Degree Fermat Fourfolds*” . ResearchGate preprint. ResearchGate
- **2026.09** | Hongmin Li. “*Low- t Coefficients of the Valley Delta Conjecture*” . ResearchGate preprint. ResearchGate
- **2026.09** | Linghu Fan, Hongmin Li. “*Resolutions of Linear p -Cyclic Quotient Singularities*” . arXiv preprint · 2609.07182 [math.AG]. arXiv
- **2026.06** | Hongmin Li. “*The Calibration Turn in AI-Assisted Research: A Conceptual and Methodological Framework for Evidence-Licensed Claims*” . arXiv preprint · 2606.31273 [cs.LG]. arXiv, Code & artifacts
- **2026.05** | Hongmin Li. “*Targeted Tests for LLM Reasoning: An Audit-Constrained Protocol*” . arXiv preprint · 2605.11599 [cs.LG]. arXiv
- **2026.05** | Hongmin Li. “*Separating Shortcut Transition from Cross-Family OOD Failure in a Minimal Model*” . arXiv preprint · 2605.12945 [cs.LG]. arXiv
- **2026.05** | Hongmin Li. “*A Controlled Counterexample to Strong Proxy-Based Explanations of OOD Performance: in a Fixed Pretraining-and-Probing Setup*” . arXiv preprint · 2605.11554 [cs.LG]. arXiv
- **2026.05** | Hongmin Li. “*FastUMAP: Scalable Dimensionality Reduction via Bipartite Landmark Sampling*” . arXiv preprint · 2605.11428 [cs.LG]. arXiv
- **2025.12** | Bian Bian, Yiming Zhang, Hongmin Li, Jiuzhou Zhong, Yutaka Saito. “*Large generative mRNA language foundation model for efficient coding sequence generation and design with mRNA-GPT*” . bioRxiv. Preprint
- **2023.9** | Hao Wu, Xiangjun Pei, Takashi Oguchi, Hongmin Li, Zifei Wang, Zhihao He, Runqiu Huang. “*An Integrated Physical Approach to Earthquake-Induced Landslide Susceptibility Incorporating Geological Structure: A Case Study of the Diexi Catchment, Sichuan, China*” . SSRN preprint. SSRN

Journal Articles

- **2026.09** | Hongmin Li, Goro Terai, Takumi Otagaki, Kiyoshi Asai. “*Gradient-based Optimization for mRNA Sequence Design*” . Bioinformatics. Journal / DOI, Preprint, Code
- **2023.1** | Hongmin Li, Xiucui Ye, Akira Imakura, Tetsuya Sakurai. “*LSEC: Large-scale spectral ensemble clustering*” . Intelligent Data Analysis. arXiv, Code
- **2022.8** | Hongmin Li, Xiucui Ye, Akira Imakura, Tetsuya Sakurai. “*Divide-and-conquer based Large-Scale Spectral Clustering*” . Neurocomputing. Preprint, Code
- **2022.7** | Meng Huang, Xiucui Ye, Hongmin Li, Tetsuya Sakurai. “*Missing Value Imputation With Low-Rank Matrix Completion in Single-Cell RNA-Seq Data by Considering Cell Heterogeneity*” . Frontiers in Genetics. DOI
- **2020.7** | Xiucui Ye, Hongmin Li, Akira Imakura, Tetsuya Sakurai. “*An Oversampling Framework for Imbalanced Classification Based on Laplacian Eigenmaps*” . Neurocomputing. Link
- **2019** | Xiucui Ye, Hongmin Li, Tetsuya Sakurai, Pei-Wei Shueng. “*Ensemble Feature Learning to Identify Risk Factors for Predicting Secondary Cancer*” . International Journal of Medical Sciences. DOI

Conference Papers

- **2020.11** | Hongmin Li, Xiucan Ye, Akira Imakura, Tetsuya Sakurai. “*Ensemble Learning for Spectral Clustering*” . ICDM 2020. PDF, Code
 - **2020.7** | Hongmin Li, Xiucan Ye, Akira Imakura, Tetsuya Sakurai. “*Hubness-based Sampling Method for Nyström Spectral Clustering*” . IJCNN 2020. Link
 - **2019.8** | Xiucan Ye, Hongmin Li, Akira Imakura, Tetsuya Sakurai. “*Distributed Collaborative Feature Selection Based on Intermediate Representation*” . IJCAI 2019. Link
 - **2018.10** | Xiucan Ye, Hongmin Li, Tetsuya Sakurai, Zhi Liu. “*Large Scale Spectral Clustering Using Sparse Representation Based on Hubness*” . CBDCOM 2018. Link, Code
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Conference presentations

- **2026.06.09** | **Poster presentation** —JST CREST BioDX Area Meeting —5th area meeting · Numazu, Japan
 - **2025.11** | **Poster presentation** —JST CREST BioDX Interim Symposium —Event
 - **2025.8** | **Oral presentation** —Input Data Differentiable Designer (ID3) at RNA Informatics Dojo 2025 — Early presentation of the method later developed into the preprint “Gradient-based Optimization for mRNA Sequence Design” ; Event, Paper, Code
 - **2024.10** | **Poster presentation** —FastUMAP at the Asia-Pacific Bioinformatics Joint Conference 2024 —Early conference presentation of the work later developed into the FastUMAP preprint; Event, Paper
 - **2024.8** | **Oral presentation** —RNA Informatics Dojo 2024
 - **2024.8** | **Participation** —IJCAI 24 (Jeju)
 - **2024.6** | **Participation** —WCCI 24 (Yokohama)
 - **2024.6** | **Participation** —25th Annual Meeting of the RNA Society of Japan
 - **2023.8** | **Oral presentation** —RNA Informatics Dojo 2023
 - **2022.8** | **Oral presentation** —RNA Informatics Dojo 2022
 - **2022.3** | **Presentation** —JST SPRING recipients event
 - **2021.12** | **Oral presentation** —CollaboTICS 2021
 - **2019.6** | **Poster presentation** —G20 Digital Science Now Symposium
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Peer Review Activities

- **2026.07** | Conference on Neural Information Processing Systems (NeurIPS)
 - **2026.07** | Neurocomputing
 - **2026.05** | Journal of King Saud University - Computer and Information Sciences
 - **2026.01** | The Journal of Supercomputing
 - **2025** | International Journal of Machine Learning and Cybernetics
 - **2024.11** | Pattern Analysis and Applications
 - **2024.11** | Knowledge and Information Systems
 - **2024.10** | Neurocomputing
 - **2024.09** | The Visual Computer
 - **2024.03** | International Journal of Machine Learning and Cybernetics
 - **2022.11** | Transactions on Pattern Analysis and Machine Intelligence
 - **2022.10** | Briefings in Functional Genomics
 - **2021–2022** | Pattern Recognition
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Honors & fellowships

- **2021.10** | **Doctoral research support** —JST SPRING —Support for Pioneering Research Initiated by the Next Generation

- **2020.10 | AETA Earthquake Prediction AI Algorithm Competition 2019 —Second Prize** —Second-place recognition for an AI model predicting earthquakes from real-time monitoring data · awarded in 2020; News, Slides
 - **2019.10 | 3rd Analysys International Algorithm Competition —Special Award** —Special recognition in the website page-view and unique-visitor prediction task
 - **2018.10 | IEEE International Conference on Cloud and Big Data Computing 2018 —Best Paper Award** —Awarded for “Large Scale Spectral Clustering Using Sparse Representation Based on Hubness”
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Skills

Programming Languages

- Python, R, MATLAB, C/C++, Java

Machine Learning & Deep Learning

- PyTorch, TensorFlow, scikit-learn
- Spectral Clustering, Large Language Models
- Bioinformatics Applications

Computational Biology

- RNA Sequence Analysis
- mRNA Design and Optimization
- Single-cell RNA-seq Data Analysis